

Student Name: _____

Maths class: _____



James Ruse Agricultural High School

2022

YEAR 12 Task 3 Examination

Mathematics Extension 2

General Instructions

Total marks: 40

- Reading time – 5 minutes
- Working time – 90 min
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 6–10, show relevant mathematical reasoning and/ or calculations

Section I – 5 marks

- Attempt Questions 1–5
- Allow about 7 minutes for this section

Section II – 60 marks

- Attempt Questions 6–10
- Allow about 83 minutes for this section

Section I

5 marks

Attempt Questions 1–5

Allow about 7 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1. Vector $\vec{u} = 3\vec{i} - 5\vec{j} + 4\vec{k}$, and $\vec{u} + \vec{v} = \vec{0}$. Which of the following is the unit vector of $\vec{v}$?

(A) $-\frac{3\sqrt{2}}{10}\vec{i} + \frac{\sqrt{2}}{2}\vec{j} - \frac{4\sqrt{2}}{5}\vec{k}$

(B) $\frac{3\sqrt{2}}{10}\vec{i} - \frac{\sqrt{2}}{2}\vec{j} + \frac{4\sqrt{2}}{5}\vec{k}$

(C) $-\frac{3\sqrt{2}}{10}\vec{i} + \frac{\sqrt{2}}{2}\vec{j} - \frac{2\sqrt{2}}{5}\vec{k}$

(D) $\frac{3\sqrt{2}}{10}\vec{i} - \frac{\sqrt{2}}{2}\vec{j} + \frac{2\sqrt{2}}{5}\vec{k}$

2. Which of the following vector equations represents a line passing through the point $A(-7, 1, 3)$ and is perpendicular to the vector $\vec{d} = -\vec{i} + 2\vec{j} - 4\vec{k}$?

(A) $\vec{r} = \begin{pmatrix} -7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}$

(B) $\vec{r} = \begin{pmatrix} -7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$

(C) $\vec{r} = \begin{pmatrix} -7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$

(D) $\vec{r} = \begin{pmatrix} -7 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$

3. How many of the following vector equations is the same as the Cartesian equation $3x + 2y = 6$?

I. $\vec{r} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \end{pmatrix}$ for $\lambda \in \mathbb{R}$

III. $\vec{r} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ for $\lambda \in \mathbb{R}$

II. $\vec{r} = \begin{pmatrix} 1 \\ 1.5 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ for $\lambda \in \mathbb{R}$

IV. $\vec{r} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 6 \end{pmatrix}$ for $\lambda \in \mathbb{R}$

(A) 0

(B) 1

(C) 2

(D) 3

4. For $3^n < n!$ where $n \in \mathbb{Z}^+$, the statement is true for

(A) $n \geq 7$

(B) $n \geq 6$

(C) $n \geq 3$

(D) All $n \in \mathbb{Z}^+$

5. If a and b are real and c and d are imaginary, which of the following inequalities will be true.

(A) $a^2c^2 + b^2d^2 \geq 2abcd$ and $a^2b^2 + c^2d^2 \geq 2abcd$

(B) $a^2c^2 + b^2d^2 \leq 2abcd$ and $a^2b^2 + c^2d^2 \geq 2abcd$

(C) $a^2c^2 + b^2d^2 \leq 2abcd$ and $a^2b^2 + c^2d^2 \leq 2abcd$

(D) $a^2c^2 + b^2d^2 \geq 2abcd$ and $a^2b^2 + c^2d^2 \leq 2abcd$

Section II begins on the next page

Section II

60 marks

Attempt Questions 6-12

Allow about 83 minutes for this section

Start each question on a new page. Extra paper is available.

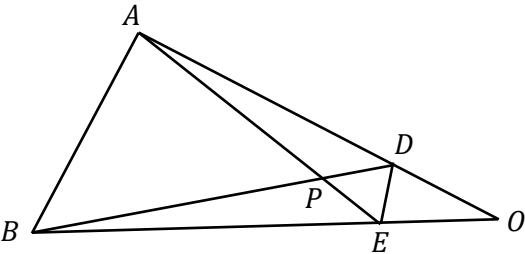
In Questions 6-10, your responses should include relevant mathematical reasoning and/ or calculations.

Question 6 (12 marks) Start a new page.

(a)	Determine whether the point $Q(-3, 1, 4)$ lies on the line $\tilde{r} = \begin{pmatrix} 6 \\ 1 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix}$, giving reasons.	2
(b)	Let P, A and B be the points $(-3, 1, -2)$, $(-4, 2, -3)$ and $(-2, 3, -1)$, respectively. (i) Find $\text{proj}_{\tilde{a}} \tilde{b}$, where $\overrightarrow{PA} = \tilde{a}$, $\overrightarrow{PB} = \tilde{b}$. (ii) Find the perpendicular distance from the point B to the line PA .	2 2
(c)	The point $P(x, y, z)$ lies on the sphere of radius 1 centred at the origin O . Prove that $ xy + yz + xz \leq 1$.	3
(d)	Use Mathematical Induction to prove that, for $n \geq 1$ $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} < \frac{3}{2}.$	3

End of Question 6.

Question 7 (12 marks) Start a new page.

(a)	Determine whether it is possible to find numbers p and q which satisfy the vector equation $p \begin{pmatrix} 2 \\ -4 \\ 3 \end{pmatrix} - q \begin{pmatrix} -3 \\ 7 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 8 \end{pmatrix}.$	3
(b)	In $\triangle ABO$ shown below, let $\overrightarrow{OD} = \tilde{a}$, $\overrightarrow{OE} = \tilde{b}$.  It is given that $\overrightarrow{DA} = 3\overrightarrow{OD}$ and $\overrightarrow{EB} = 4\overrightarrow{OE}$. (i) Prove that $\overrightarrow{AB} = 5\tilde{b} - 4\tilde{a}$. (ii) If $\overrightarrow{AB} = \overrightarrow{AD}$ and $\overrightarrow{OD} = \overrightarrow{OE}$, use vector methods to find the value of $\cos(\angle AOB)$. AE and BD intersect at the point P (iii) Find a vector equation of line AE. (iv) Find a vector equation of line BD. (v) Find $\overrightarrow{OP}$ in terms of $\tilde{a}$ and $\tilde{b}$.	1 3 1 1 3

End of Question 7.

Question 8 (12 marks) Start a new page.

(a)	(i) Prove for integers $n > r > 1$, $\binom{n}{r} < n \binom{n-1}{r-1}$. (ii) Given that $(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n$ show that for $a, b > 0$ and integers $n > 1$, $(a+b)^n - a^n < nb(a+b)^{n-1}$.	2 2
(b)	Suppose $0 \leq t \leq \frac{1}{\sqrt{2}}$. (i) Show that $0 \leq \frac{2t^2}{1-t^2} \leq 4t^2$. (ii) Hence show that $0 \leq \frac{1}{1+t} + \frac{1}{1-t} - 2 \leq 4t^2$. (iii) By integrating the result in part (ii) from $t = 0$ to $t = x$ (where $0 \leq x \leq \frac{1}{\sqrt{2}}$), show that $0 \leq \log_e \left(\frac{1+x}{1-x} \right) - 2x \leq \frac{4x^3}{3}.$ (iv) Hence show that for $0 \leq x \leq \frac{1}{\sqrt{2}}$, $1 \leq \left(\frac{1+x}{1-x} \right) e^{-2x} \leq e^{\frac{4x^3}{3}}$.	2 2 2 2

End of Question 8.

Question 9 (12 marks) **Start a new page.**

(a)	Let $a + b + c = 1$ where a, b , and c are positive real numbers. Find the smallest possible value of $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$.	3
(b)	(i) Let a be a positive real number. Show that $a + \frac{1}{a} \geq 2$. (ii) If $a^n + \frac{1}{a^n} \geq a^{n-1} + \frac{1}{a^{n-1}}$ for all positive integers $n \geq 1$, prove $a^{n+1} + \frac{1}{a^{n+1}} \geq a^n + \frac{1}{a^n}$. (iii) Let n be a positive integer and $a_1, a_2, \dots, a_n$ be n positive real number. Prove by induction that $(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right) \geq n^2$. (iv) Hence show that $\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta \geq 9$.	1 2 4 2

End of Question 9.

Question 10 (12 marks) **Start a new page.**

(a)	The function f is defined by $f(x) = \frac{1}{2}(1 - \sqrt{1-x})$ for $0 < x < 1$. Given that $f_1 = f$ and $f_n = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ compositions}}$, prove by Mathematical Induction that for $0 < \theta < \frac{\pi}{2}$ and positive integers n $f_n(\sin^2 \theta) = \sin^2 \left(\frac{\theta}{2^n} \right).$	5
(b)	In three-dimensional space, the origin is O. The sphere S has its centre at the point C, where $\overrightarrow{OC} = \underline{\underline{c}}$. The radius of the sphere is r, and the position vector of a point on the sphere is $\underline{\underline{m}}$. It is given that $\underline{\underline{m}} - \underline{\underline{c}} = r$. The line ℓ is tangent to the sphere S at the point $\underline{\underline{p}}$ and is parallel to the vector $\underline{\underline{n}}$. (i) State a vector equation of the line ℓ. (ii) Prove that $(\underline{\underline{p}} - \underline{\underline{c}}) \cdot \underline{\underline{n}} = 0$, giving reasons. (iii) Show that $\lambda = \frac{\sqrt{r^2 - \underline{\underline{p}} - \underline{\underline{c}} ^2}}{ \underline{\underline{n}} }$.	1 2 4

END of PAPER

Multiple choice:

1. C
2. D
3. B
4. A
5. B

Year 12 Extension 2: Q6		
Suggested Solutions	Marks Awarded	Marker's Comments
(a) The point $Q(-3,1,4)$ lies on the line $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 3 \\ -2 \end{bmatrix}$ iff $\exists \lambda$ such that $\begin{bmatrix} -3 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \\ -5 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 3 \\ -2 \end{bmatrix}$ By row 1, $-3 = 6 + 3\lambda \rightarrow \lambda = -3$ We check this parameter value with row 2 to find $1 \neq 1 + (-3)(3)$ Hence the point Q does NOT lie on the line. (b) Let $P = (-3,1,-2), \quad A = (-4,2,-3), \quad B = (-2,3,-1)$ (i) We have: $\overrightarrow{PA} = \begin{bmatrix} -4 - (-3) \\ 2 - 1 \\ -3 - (-2) \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} := \mathbf{a}$ and $\overrightarrow{PB} = \begin{bmatrix} -2 - (-3) \\ 3 - 1 \\ -1 - (-2) \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} := \mathbf{b}$ Then $\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} ^2} \mathbf{b}$ Now $\mathbf{a} \cdot \mathbf{b} = (-1)(1) + (1)(2) + (-1)(1) = 0$, which means $\text{proj}_{\mathbf{a}} \mathbf{b} = 0\mathbf{b} = \mathbf{0}$ The projection is the zero vector.	1 First mark for identifying what had to be done (i.e. parameter consistency check). 1 Final mark for showing at least two equations were inconsistent relative to λ and its consequence. 1 First mark for obtaining both vectors correctly. 1 Final mark for showing the projection was the zero vector (not scalar).	

(ii) Since $\overrightarrow{PB} \cdot \overrightarrow{PA} = 0$, the line segment PB must be perpendicular to the line through PA. Hence the perpendicular distance is simply the magnitude of $\mathbf{b}$: $ \mathbf{b} = \sqrt{(1)^2 + (2)^2 + (1)^2} = \sqrt{6}$ units. (c) Consider that $0 \leq (x - y)^2 + (y - z)^2 + (z - x)^2$ $= 2(x^2 + y^2 + z^2) - 2(x y + y z + z x)$ $= 2 \cdot 1 - 2(xy + yz + zx)$ where the last line follows as $x^2 + y^2 + z^2 = 1$. Rearranging, we have $ xy + yz + zx \leq 1$	1 First mark for justification (i.e. the line segments are at right angles as a result of the dot product being 0. 1 Final mark for correct distance. 1 First mark for appropriate setup. 1 One mark for correct use of the restriction $x^2 + y^2 + z^2 = 1$. 1 Final mark for correct conclusion.	If non-standard inequalities (i.e. those not listed in the syllabus) were used without proof, even if true, full marks were not awarded.
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(d) We need to show that $S(n): 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^n} < 2$ is the case $\forall n \in \mathbb{N}$. Consider now that $1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n} + \frac{1}{2^{n+1}} = 1 + \frac{1}{2} \left(1 + \frac{1}{2} + \cdots + \frac{1}{2^n} \right)$ $< 1 + \frac{1}{2}(2) \quad \text{if } S(n) \text{ is true}$ $= 2$ Hence, if the claim is true for case n, it follows it's true for case $n + 1$. Now, for $n = 1$, we have $\text{LHS} = 1 + \frac{1}{2} = \frac{3}{2} < 2 = \text{RHS}$ Since the claim is true for $n = 1$, and since it is true for $n + 1$ if true for n, it follows that the claim is true for all integers $n \geq 1$.	1 First mark for testing of a base case. 1 Second mark for correct use of hypothesis in the proof (which includes stating its use when used). 1 Final mark for correct, logical conclusion.	Many students used sum of geometric series: $1 + \frac{1}{2} + \cdots + \frac{1}{2^n} = 2 - \frac{1}{2^n}$ as a starting point, but if the induction proof had been worked properly, this path would have led nowhere helpful. If the sum is used, then you're forced to take as hypothesis $2 - \frac{1}{2^n} < 2$ as true. Next step would be to show that if this is true, then $1 + \frac{1}{2} + \cdots + \frac{1}{2^n} + \frac{1}{2^{n+1}} = \left(2 - \frac{1}{2^n} \right) + \frac{1}{2^{n+1}} < 2 + \frac{1}{2^{n+1}} \quad (\text{by hyp.})$ which isn't less than 2. Using the geometric sum doesn't help. It also makes the fact that the sum is less than 2 so obvious that working with it gets in the way.
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MATHEMATICS Extension 2: Question...7...

Suggested Solutions	Marks	Marker's Comments
a) $2p + 3q = -2$ - ① $-4p - 7q = 2$ - ② $3p - 2q = 8$ - ③ from ①: $q = \frac{-2-2p}{3}$ sub. q into ②: $-4p - 7\left(\frac{-2-2p}{3}\right) = 2$ $-12p + 14 + 14p = 6$ $2p = -8$ $p = -4$ $\therefore q = 2$ sub. $p = -4$ and $q = 2$ into ③ $\text{LHS} = 3(-4) - 2(2)$ $= -16 \neq 8$ $\therefore$ no possible p and q	1 1 1	show the equation for the z component as well. there are other values for p and q need to have conclusion
b) i) $\vec{AB} = \vec{AO} + \vec{OB}$ $= \vec{AD} + \vec{DO} + \vec{OE} + \vec{EB}$ $= -3\mathbf{a} - \mathbf{a} + \mathbf{b} + 4\mathbf{b}$ $= 5\mathbf{b} - 4\mathbf{a}$ ii) $\vec{AB} = \vec{AD}$ $ 5\mathbf{b} - 4\mathbf{a} = 3\mathbf{a} $ $(5\mathbf{b} - 4\mathbf{a})^2 = (3\mathbf{a})^2$ $25\mathbf{b} \cdot \mathbf{b} + 16\mathbf{a} \cdot \mathbf{a} - 40\mathbf{a} \cdot \mathbf{b} = 9\mathbf{a} \cdot \mathbf{a}$ $25 \mathbf{b} ^2 + 16 \mathbf{a} ^2 - 40\mathbf{a} \cdot \mathbf{b} = 9 \mathbf{a} ^2$	1 1	Starting with $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{ \mathbf{a} \mathbf{b} }$ leads nowhere and hence received 0 marks

Suggested Solutions	Marks	Marker's Comments
as $\vec{OD} = \vec{OE}$ $\underline{a} = \underline{b}$ $\therefore \underline{a} ^2 = \underline{b} ^2$ $\therefore 25 \underline{a} ^2 + 16 \underline{a} ^2 - 40\underline{a} \cdot \underline{b} = 9 \underline{a} ^2$ $32 \underline{a} ^2 = 40\underline{a} \cdot \underline{b}$ $\frac{\underline{a} \cdot \underline{b}}{ \underline{a} ^2} = \frac{32}{40}$ $\frac{\underline{a} \cdot \underline{b}}{ \underline{a} \underline{b} } = \frac{4}{5}$ $\therefore \cos \angle AOB = \frac{4}{5}$ iii) line AE: $\underline{r} = \underline{b} + \lambda(\underline{b} - 4\underline{a})$ $\lambda \in \mathbb{R}$ or $\underline{r} = 4\underline{a} + \lambda(\underline{b} - 4\underline{a})$ iv) line BD: $\underline{r} = \underline{a} + \mu(\underline{a} - 5\underline{b})$ $\mu \in \mathbb{R}$ or $\underline{r} = 5\underline{b} + \mu(\underline{a} - 5\underline{b})$ v) P is the intersection of AE and BD $\underline{b} + \lambda(\underline{b} - 4\underline{a}) = \underline{a} + \mu(\underline{a} - 5\underline{b})$ $-4\lambda\underline{a} + (1+\lambda)\underline{b} = (1+\mu)\underline{a} - 5\mu\underline{b}$ equating coefficients: $-4\lambda = 1+\mu \quad - \textcircled{1}$ $1+\lambda = -5\mu$ $\lambda = -1-5\mu \quad - \textcircled{2}$ sub. $\textcircled{2}$ into $\textcircled{1}$	1 1 1 1 1	

MATHEMATICS Extension 2: Question...7...

Suggested Solutions	Marks	Marker's Comments
$-4(-1-5\mu) = 1+\mu$ $4+20\mu = 1+\mu$ $19\mu = -3$ $\mu = -\frac{3}{19}$ $\therefore \vec{OP} = \underline{a} + \left(-\frac{3}{19}\right)(\underline{a} - 5\underline{b})$ $= \frac{16}{19}\underline{a} + \frac{15}{19}\underline{b}$	1 1	

Question 8 :

(a) (i) $n > r > 1, \binom{n}{r} < n \binom{n-1}{r-1}$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n}{r} \times \frac{(n-1)!}{(r-1)!(n-r)!}$$

$$= \frac{n}{r} \binom{n-1}{r-1} \quad \checkmark$$

$$r > 1 \therefore \frac{1}{r} < 1 \quad \checkmark$$

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1} < n \binom{n-1}{r-1} \quad \checkmark$$

(ii) $(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \dots + \binom{n}{n}b^n$

$$(a+b)^n - a^n = \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n \quad \checkmark$$

$$\leq n \binom{n-1}{0}a^{n-1}b + n \binom{n-1}{1}a^{n-2}b^2 + \dots + n \binom{n-1}{n-1}b^n$$

$$= nb \left[\binom{n-1}{0}a^{n-1} + \binom{n-1}{1}a^{n-2}b + \dots + \binom{n-1}{n-1}b^{n-1} \right] \quad \checkmark$$

$$= nb (a+b)^{n-1} \quad \checkmark$$

$$(8) \quad 0 < t \leq \frac{1}{\sqrt{2}}$$

$$(i) \quad 0 < t \leq \frac{1}{\sqrt{2}} \quad \therefore 0 < t^2 \leq \frac{1}{2} \quad \therefore 0 < 2t^2 \leq 1$$

$$0 < t^2 \leq \frac{1}{2} \quad \therefore -\frac{1}{2} \leq -t^2 \leq 0 \quad \therefore \frac{1}{2} \leq 1-t^2 \leq 1$$

$$\therefore 1 \leq \frac{1}{1-t^2} \leq 2 \quad \checkmark$$

$$\therefore 2t^2 \leq \frac{2t^2}{1-t^2} < 4t^2 \quad \checkmark$$

$$\therefore 0 \leq \frac{2t^2}{1-t^2} \leq 4t^2 \quad \text{since } 2t^2 > 0 \quad \checkmark$$

$$(ii) \quad \frac{2t^2}{1-t^2} = \frac{2t^2 - 2 + 2}{1-t^2} = \frac{2(t^2 - 1)}{1-t^2} + \frac{2}{1-t^2}$$

$$= -2 + \frac{2}{1-t^2} \quad \checkmark$$

$$= -2 + \frac{1}{1-t} + \frac{1}{1+t} \quad \checkmark$$

$$\therefore 0 \leq \frac{1}{1+t} + \frac{1}{1-t} - 2 \leq 4t^2$$

$$(iii) \quad \int_0^x 0 \, dt \leq \int_0^x \left(\frac{1}{1+t} + \frac{1}{1-t} - 2 \right) dt \leq \int_0^x 4t^2 \, dt$$

$$0 \leq \left[\ln|1+t| - \ln|1-t| - 2t \right]_0^x \leq \left[\frac{4t^3}{3} \right]_0^x \quad \checkmark$$

$$0 \leq \ln \left(\frac{1+x}{1-x} \right) - 2x \leq \frac{4x^3}{3} \quad \checkmark \quad \frac{1+x}{1-x} > 0$$

$$ii) \quad 0 \leq \log_e \left(\frac{1+x}{1-x} \right) - 2x \leq \frac{4x^3}{3}$$

$$e^0 \leq e^{\log_e \left(\frac{1+x}{1-x} \right) - 2x} \leq e^{\frac{4x^3}{3}} \quad \checkmark$$

$$1 \leq e^{\log_e \left(\frac{1+x}{1-x} \right)} e^{-2x} \leq e^{\frac{4x^3}{3}}$$

$$1 \leq \left(\frac{1+x}{1-x} \right) e^{-2x} \leq e^{\frac{4x^3}{3}} \quad \checkmark$$

Question 9

Let $a + b + c = 1$ where a, b , and c are positive real numbers. Find the smallest possible value of $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$.	3
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a) $(\sqrt{x} - \sqrt{y})^2 \geq 0$
 $x - 2\sqrt{xy} + y \geq 0$
 $\therefore x + y \geq 2\sqrt{xy}$

$\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$
 $= \frac{(1-a)(1-b)(1-c)}{abc}$
 $= \frac{(b+c)(a+c)(a+b)}{abc}$ since $a+b+c=1$ ① using $a+b+c=1$ following some helpful manipulation of $\left(\frac{1}{a}-1\right)\left(\frac{1}{b}-1\right)\left(\frac{1}{c}-1\right)$
 $\geq \frac{2\sqrt{bc} \times 2\sqrt{ac} \times 2\sqrt{ab}}{abc}$ since $x+y \geq 2\sqrt{xy}$ ① using AM-GM inequality
 $= \frac{8abc}{abc}$
 $= 8$
 $\therefore$ Smallest possible value is 8. ① answer

① mark for using $a+b+c=1$ following some helpful manipulation of $\left(\frac{1}{a}-1\right)\left(\frac{1}{b}-1\right)\left(\frac{1}{c}-1\right)$

①

b) (i) Let a be a positive real number. Show that $a + \frac{1}{a} \geq 2$.

$(a-1)^2 \geq 0$ OR $(\sqrt{a} - \frac{1}{\sqrt{a}})^2 \geq 0$ ①
 $a^2 - 2a + 1 \geq 0$
 $a^2 + 1 \geq 2a$
 $a + \frac{1}{a} \geq 2$
 $a - 2 + \frac{1}{a} \geq 0$
 $a + \frac{1}{a} \geq 2$

(ii) If $a^n + \frac{1}{a^n} \geq a^{n-1} + \frac{1}{a^{n-1}}$ for all positive integers $n \geq 1$, prove $a^{n+1} + \frac{1}{a^{n+1}} \geq a^n + \frac{1}{a^n}$.

Given

$a^{n+1} + \frac{1}{a^{n+1}} \geq a^{n+1} + \frac{1}{a^{n+1}}$

we can map n to $n+1$:

$a^{n+1} + \frac{1}{a^{n+1}} \geq a^{n+1} + \frac{1}{a^{n+1}}$ ②

(iii) Let n be a positive integer and $a_1, a_2, \dots, a_n$ be n positive real number. Prove by induction that $(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right) \geq n^2$.

(iii) Prove true for $n=1$

When $n=1$

LHS = $a_1 \times \frac{1}{a_1} = 1$

RHS = $1^2 = 1 = \text{LHS}$

$\therefore$ True for $n=1$ ①

Assume true for $n=k$

ie assume

$(a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}\right) \geq k^2$

Prove true for $n=k+1$

ie. prove

$(a_1 + a_2 + \dots + a_k + a_{k+1}) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k} + \frac{1}{a_{k+1}}\right) \geq (k+1)^2$

LHS = $(\underbrace{a_1 + a_2 + \dots + a_k}_{a_k} + a_{k+1}) \left(\underbrace{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}}_{\frac{1}{a_k}} + \frac{1}{a_{k+1}}\right)$

$= (a_1 + a_2 + \dots + a_k) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}\right) + \frac{1}{a_{k+1}} (a_1 + a_2 + \dots + a_k) + a_{k+1} \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}\right) + \left(\frac{1}{a_{k+1}}\right) a_{k+1}$

$\geq k^2 + \frac{1}{a_{k+1}} (a_1 + a_2 + \dots + a_k) + a_{k+1} \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_k}\right) + 1$ ① by assumption

$= k^2 + \sum_{j=1}^k \left(\frac{a_j}{a_{k+1}} + \frac{1}{a_{k+1}}\right) + 1$

$\geq k^2 + \sum_{j=1}^k 2 + 1$ (from (i)) ①

$= k^2 + 2k + 1$

$= (k+1)^2$

$= \text{RHS}$

$\therefore$ True for $n=k+1$ + conclusion

(iv) Hence show that $\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta \geq 9$.

Let $a_1 = \cos^2 \theta$, $a_2 = \sin^2 \theta$, $a_3 = \tan^2 \theta$ ①

$(\sin^2 \theta + \cos^2 \theta + \tan^2 \theta) (\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) \geq 3^2$ from (iii) need to state this

$(1 + \tan^2 \theta) (\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) \geq 9$

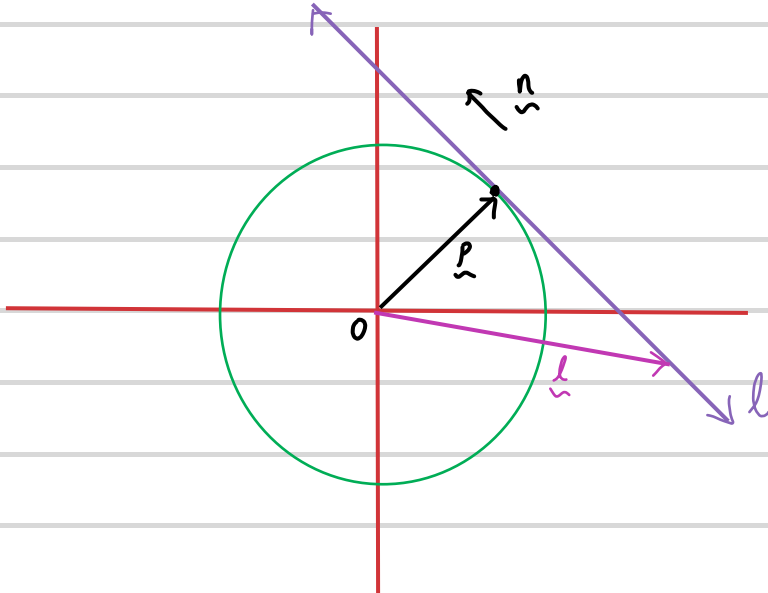
$\sec^2 \theta (\operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta) \geq 9$

$\therefore \operatorname{cosec}^2 \theta + \sec^2 \theta + \cot^2 \theta \geq 9 \cos^2 \theta$ ①

Suggested Solutions	Marks	Marker's Comments
a) Test for $n=1$ $ \begin{aligned} f_1(\sin^2 \theta) &= \frac{1}{2}(1 - \sqrt{1 - \sin^2 \theta}) \\ &= \frac{1}{2}(1 - \sqrt{\cos^2 \theta}) \\ &= \frac{1}{2}(1 - \cos \theta) \\ &= \frac{1}{2}(1 - \cos \theta) \quad (\text{since } 0 < \theta < \frac{\pi}{2}) \\ &= \frac{1}{2}(1 - (1 - 2\sin^2 \frac{\theta}{2})) \\ &= \sin^2 \frac{\theta}{2} \end{aligned} $ $\therefore$ The statement is true for $n=1$ Assume the statement is true for $n=k$, $k \in \mathbb{Z}$, $k \geq 1$ $\therefore f_k(\sin^2 \theta) = \sin^2 \left(\frac{\theta}{2^k} \right)$ Test for $n=k+1$, RTP: $f_{k+1}(\sin^2 \theta) = \sin^2 \left(\frac{\theta}{2^{k+1}} \right)$ $ \begin{aligned} \text{LHS} &= f_{k+1}(\sin^2 \theta) \\ &= f \circ f_k(\sin^2 \theta) \\ &= f\left(\sin^2 \left(\frac{\theta}{2^k} \right)\right) \quad (\text{By assumption}) \\ &= \frac{1}{2} \left(1 - \sqrt{1 - \sin^2 \left(\frac{\theta}{2^k} \right)} \right) \\ &= \frac{1}{2} \left(1 - \sqrt{\cos^2 \left(\frac{\theta}{2^k} \right)} \right) \\ &= \frac{1}{2} \left(1 - \left \cos \left(\frac{\theta}{2^k} \right) \right \right) \\ &= \frac{1}{2} \left(1 - \cos \left(\frac{\theta}{2^k} \right) \right) \quad (0 < \frac{\theta}{2^k} < \frac{\pi}{2}) \\ &= \frac{1}{2} \left(1 - (1 - 2\sin^2 \left(\frac{\theta}{2^{k+1}} \right)) \right) \\ &= \frac{1}{2} (2\sin^2 \left(\frac{\theta}{2^{k+1}} \right)) \\ &= \sin^2 \left(\frac{\theta}{2^{k+1}} \right) \end{aligned} $ $\therefore$ If the statement is true for $n=k$, then it is true for $n=k+1$, $\therefore$ The statement is true by the principle of mathematical induction	① ① ① ① ①	Explain why the absolute value is not needed using appropriate trig identity without skipping steps correct assumption Substitution with the phrase "assumption" Finish the proof off correctly with reason.

Suggested Solutions	Marks	Marker's Comments
b) i. $\underline{l} : \underline{l} = \underline{p} + \lambda \underline{n}$	1	
ii Since $\underline{p} - \underline{c}$ represents the vector from the centre to $\underline{p}$, then it is perpendicular to $\underline{l}$. (radius is perpendicular to the tangent at the point of contact) Since $\underline{l}$ has direction vector $\underline{n}$, then $\underline{p} - \underline{c}$ is perpendicular to $\underline{n}$. $\therefore (\underline{p} - \underline{c}) \cdot \underline{n} = 0$	1	circle geometry property
iii Since $\underline{p}$ lies on $\underline{l}$ and the surface of the sphere: Solving $\underline{l} = \underline{p} + \lambda \underline{n}$ and $\underline{m} - \underline{c} = r$ simultaneously, $(\underline{l} + \lambda \underline{n}) - \underline{c} = r$ (substitute $\underline{l}$ into $\underline{m}$) $(\underline{p} - \underline{c}) + \lambda \underline{n} ^2 = r^2$ $((\underline{p} - \underline{c}) + \lambda \underline{n}) \cdot ((\underline{p} - \underline{c}) + \lambda \underline{n}) = r^2$ $\underline{p} - \underline{c} ^2 + 2\lambda(\underline{p} - \underline{c}) \cdot \underline{n} + \lambda^2 \underline{n} ^2 = r^2$ $\therefore \underline{p} - \underline{c} ^2 + \lambda^2 \underline{n} ^2 = r^2$ ($(\underline{p} - \underline{c}) \cdot \underline{n} = 0$ was proven in ii). $\therefore \lambda^2 = \frac{r^2 - \underline{p} - \underline{c} ^2}{ \underline{n} ^2}$ $\lambda = \frac{\sqrt{r^2 - \underline{p} - \underline{c} ^2}}{ \underline{n} }$ $= 0$ (since $\underline{p} - \underline{c} = r$) <u>or</u> Since $\underline{p}$ lies on $\underline{l} = \underline{p} + \lambda \underline{n}$ (from i) then $\underline{p} = \underline{p} + \lambda \underline{n}$ $\therefore \lambda \underline{n} = \underline{0}$ $\lambda = 0$ (since $\underline{n} \neq \underline{0}$)	1	Attempt to solve simultaneously
	1	Expansion
	1	use of property in part ii
	1	Answer
	4	

Some notes about lines:



If l has equation $l: \underline{r} = \underline{p} + \lambda \underline{n}$
 $\therefore \underline{p} \perp \underline{l}$ NOT $\underline{p} \perp \underline{r}$!!!

$\underline{l}$ is the tangent, $\underline{r}$ is the variable positional vector from O to any point on the line l .